MANIPAL ACADEMY of HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

Machine Learning approach for Fraud Detection

Mini-Project Synopsis

submitted to

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MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1. INTRODUCTION

Online financial transactions have been on the rise ever since the invention of new modes of payment. Alongside the growth, credit card fraud has grown explosively, causing monthly losses in the billions. Fraudsters have been constantly upgrading their tactics, making rule-based detection systems futile. Online transaction fraud has escalated into an international crisis, giving rise to significant economic damage to financial institutions and consumers. According to the Global Anti-Scam Alliance, scammers stole around $1.03 trillion in 2024. The pace and the volume at which modern transactions take place, along with the anonymity provided by the internet, make it impossible to have manual oversight. Therefore, fraud detection has become a matter of priority and a highly focused area of research, making the model proactive and adaptive. Fraud detection has thus become a priority research area focused on creating proactive and adaptive systems. Machine learning offers a promising solution by learning complex patterns from historical transaction data and detecting anomalous behaviour indicative of fraud. Machine learning represents this essential paradigm shift, offering a dynamic, adaptive, and scalable solution to the challenges of modern fraud detection. Unlike rule-based systems that rely on explicit programming, ML models learn directly from historical data, enabling them to uncover complex, subtle, and non-intuitive patterns that are often invisible to human analysts or impossible to capture with static rules. ML algorithms, particularly ensemble methods and neural networks, can analyse vast, high-dimensional datasets containing millions of transactions. They can identify intricate correlations between hundreds of features—such as transaction amount, time of day, device fingerprint, and user behaviour to build a nuanced understanding of what constitutes "normal" versus "anomalous" activity. This allows them to detect sophisticated fraud schemes that manifest as subtle deviations from typical patterns. Perhaps the most critical advantage is the ability of ML models to adapt. As new transaction data becomes available, the models can be retrained to learn and incorporate emerging fraud tactics. This creates a dynamic defence system that evolves in tandem with the threat landscape, significantly reducing the window of vulnerability that plagues rule-based systems. Machine learning systems can process and score millions of transactions in real-time, operating at a scale and speed that is unattainable for human teams. By automating the initial screening process, ML models can significantly reduce operational costs and free up human fraud analysts to focus their expertise on the most complex and high-risk cases.

However, fraud detection presents unique challenges: extreme class imbalance makes it difficult for models to learn the minority class, and concept drift demands models that can update and adapt. Among ML techniques, ensemble methods like gradient boosting have gained prominence for fraud detection due to their high accuracy and ability to model non-linear patterns. In particular, XGBoost is an advanced boosting algorithm known for its efficiency and superior performance in classification tasks. XGBoost adds regularization and improved tree-building strategies to Gradient Boosted Trees, often outperforming single models like logistic regression or SVM. Prior studies have reported XGBoost achieving top results in fraud detection competitions and applications. XGBoost is an ensemble technique that builds a series of sequential decision trees, where each new tree corrects the errors of the previous ones. This iterative process allows the model to achieve exceptionally high accuracy and predictive power, frequently outperforming other algorithms in head-to-head comparisons on fraud detection datasets. Real-world financial data is often imperfect. XGBoost includes a sophisticated, built-in routine for handling missing values. This simplifies the data preprocessing pipeline and makes the model more robust to the types of data quality issues commonly encountered in operational datasets.

Given the high dimensionality, extreme class imbalance, and continuously evolving nature of financial transaction data, existing fraud detection systems struggle with low accuracy for the minority fraud class, high rates of false positives that disrupt legitimate customer activity, and a fundamental inability to detect novel threats in a timely manner. By leveraging XGBoost’s strengths, we aim to build a fraud detection model that can accurately classify fraudulent transactions while maintaining a low false positive rate, thereby protecting customers and reducing losses for financial institutions.

2. Literature Survey

Paper 1: "Fraud Detection in Financial Transactions Using Machine Learning Techniques" by Aman Negi and Deepak Kumar

- **Takeaway:** This paper provides a foundational understanding of why machine learning is crucial for modern fraud detection. It highlights the limitations of older, rule-based systems and effectively demonstrates the superiority of ensemble models. The key takeaway for the project is the paper's emphasis on handling class imbalance using the SMOTE (Synthetic Minority Over-sampling Technique). They show that simply applying a model to raw, imbalanced data yields poor results. By using SMOTE to create synthetic examples of the minority class which is fraud, the performance of models like XGBoost and Random Forest improved dramatically. This is a practical technique we can directly apply in the project to ensure the model learns effectively from the few fraud examples available.

- **Drawbacks:** The primary limitation is that while SMOTE is effective, it can also introduce noise into the dataset by creating synthetic samples that are not perfectly representative of real-world fraud. This can sometimes lead to the model learning slightly incorrect patterns, a risk we should be aware of. The paper also doesn't deeply explore the computational cost of these models, which can be a factor in real-time systems.

Paper 2: "Fraud Detection in Online Transaction using Hybrid CNN and XGBOOST" by P. Nagarajan and Kamali B

- **Takeaway:** This paper explores a more complex, hybrid approach by combining a Convolutional Neural Network (CNN) with a Support Vector Machine (SVM) and comparing it against XGBoost. The most important lesson from this paper is the confirmation of **X**GBoost's standalone power. Despite the complexity of the proposed hybrid model, the results show that XGBoost performs exceptionally well on its own, often rivaling or even surpassing the more intricate hybrid architecture. This validates our choice of XGBoost as a primary tool. It teaches us that starting with a powerful, well-understood algorithm like XGBoost is often a more effective strategy than immediately jumping to complex hybrid models.

- **Drawbacks:** The main drawback is the complexity and "black box" nature of the hybrid CNN-SVM model. Such a model is much harder to implement, tune, and interpret compared to XGBoost. For a team new to machine learning, building and explaining such a system would be very challenging. The paper also doesn't provide a strong justification for *why* a CNN, typically used for image processing, would be the best choice for tabular financial data.

Paper 3: "An application of XGBoost algorithm for online transaction fraud detection based on improved sailfish optimizer" by Hongwei Chen and Lun Chen

- **Takeaway:** The core lesson from this paper is the importance of **hyperparameter tuning**. It shows that the performance of a powerful algorithm like XGBoost can be further enhanced by systematically finding the best settings for its internal parameters. The key takeaway is that we should plan to experiment with different XGBoost parameters to see how they affect our model's performance. This process, even done manually or with simpler automated tools, is a critical step in building an effective model.
- **Drawbacks:** The proposed ISFO algorithm is highly complex and computationally expensive. Implementing and understanding such a metaheuristic optimizer is an advanced topic, well beyond the scope of a foundational machine learning project. The performance gains from such a complex optimizer might be marginal compared to the effort required, making it impractical for many real-world applications.

Paper 4: "Fraud Detection in Mobile Payment Systems using an XGBoost-based Framework" by Petr Hajek, Mohammed Zoynul Abedin, and Uthayasankar Sivarajah

- **Takeaway:** This paper introduces a very practical and business-oriented perspective by proposing a novel performance metric: cost savings. Instead of just focusing on abstract metrics like F1-score, it evaluates the model's effectiveness in real financial terms—how much money did the model save by preventing fraudulent transactions versus the costs it incurred. The main takeaway is to think about the real-world impact of our model. It also reinforces the effectiveness of combining a simple data-handling technique, **Random Under-Sampling (RUS)**, with XGBoost to manage class imbalance. RUS is simpler to implement than SMOTE and can be very effective, giving us another practical tool for our project.
- **Drawbacks:** The cost savings model proposed in the paper relies on several assumptions about the costs associated with fraud and false positives. In a real-world scenario, getting accurate estimates for these costs can be very difficult. Furthermore, while Random

Under-Sampling is simple, it involves throwing away a large portion of the majority class data which are legitimate transactions, which can sometimes lead to the model losing valuable information.

Paper 5: Detection of Credit Card Fraud Using Machine Learning by Vaishnavi T. et al.

- **Takeaway:** The key lesson from this paper is the effectiveness of combining feature engineering with machine learning models for fraud detection. The study demonstrates that techniques like calculating transaction distances (using the Haversine formula), extracting temporal features, and demographic analysis can significantly improve model performance. XGBoost stood out as the best-performing algorithm, achieving 99% accuracy and an AUC of 0.998, especially after hyperparameter optimization using grid search. The takeaway is that in addition to selecting a strong algorithm like XGBoost, careful feature engineering and handling of class imbalance (via scale_pos_weight) are crucial for detecting credit card fraud accurately.
- **Drawbacks:** The approach, while effective, heavily relies on feature engineering and detailed transaction data, which may not always be available for all institutions or datasets. Also, although XGBoost achieved excellent results, it is more complex to tune compared to models like Logistic Regression, which may pose challenges for beginners. The lack of explicit use of automated feature selection methods or advanced balancing techniques (such as SMOTE or ensemble strategies) could limit adaptability for other fraud detection datasets.

3. OBJECTIVES

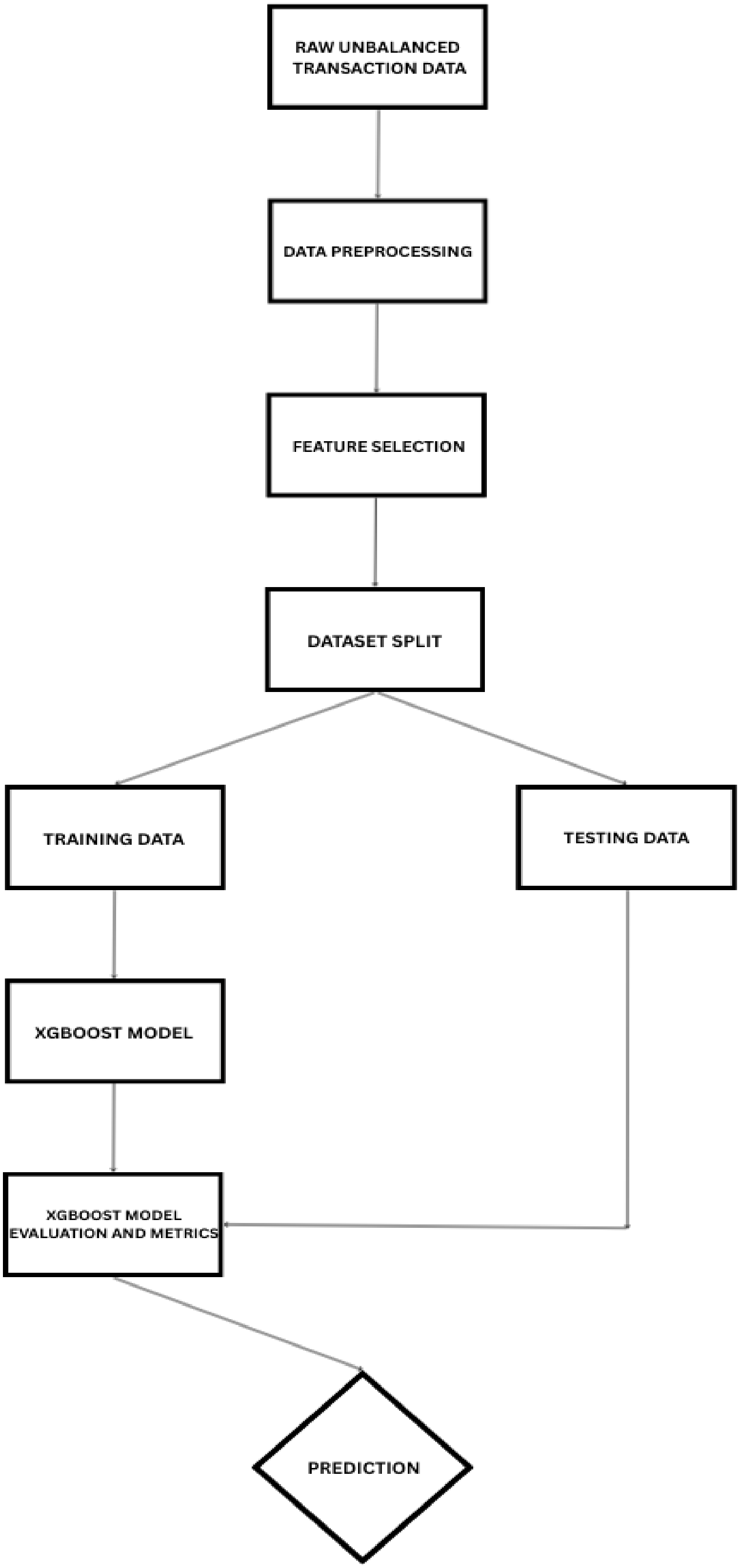
1. **To Understand and Prepare the Transaction Dataset:** The first step is always to get familiar with our data. This objective involves loading the financial transaction dataset and performing an exploratory data analysis to understand its structure, features, and statistical properties. A key part of this is to identify and visualize the severe class imbalance between fraudulent and non-fraudulent transactions.

2. **To Preprocess the Data for Model Training:** Raw data is rarely ready for a machine learning model. This objective covers the essential tasks of cleaning the data, such as handling any missing values, and encoding categorical features into a numerical format that the model can understand.

3. **To Implement and Train an XGBoost Classification Model:** This is the core technical goal of the project. We will focus on setting up the XGBoost algorithm, splitting the preprocessed data into training and testing sets, and training the model on the training data to learn the patterns that differentiate fraudulent from legitimate transactions.

4. **To Evaluate the Model's Performance Using Appropriate Metrics:** This objective is to test the trained model on the unseen test data and measure its performance. Because accuracy can be misleading in fraud detection, we will focus on more informative metrics like Precision like how many flagged transactions were fraudulent, Recall which checks if we caught most of the fraudulent transactions, and the F1-Score.

4. BLOCK DIAGRAM



5. APPLICATION

1. **Banking and Credit Card Fraud Detection:** Banks and credit card companies deploy models like XGBoost to monitor transactions in real time and flag suspicious activity on customer accounts. By analyzing attributes such as transaction amount, location, time, and past behavior, the model can detect anomalies that suggest credit card fraud or account takeover attempts. Implementing such systems helps banks prevent unauthorized charges and reduce losses; for instance, an accurate XGBoost model can avert fraudulent credit card transactions that contribute to billions in potential losses. Rapid detection also minimizes inconvenience for customers by enabling quick card blocking and fraud alerts before excessive damage occurs.

2. **E-commerce and Online Payment Gateways:** Online retailers and payment processors use fraud detection models to examine online purchases and digital payments. These systems protect against fraudulent orders, which if unchecked can result in merchandise loss and costly chargebacks for merchants. By scoring each transaction for fraud risk, the XGBoost-based detector can decline high-risk transactions. This application is vital for e-commerce companies, as it filters out fraudulent purchases while allowing legitimate customer orders to proceed without friction. Ultimately, such models reduce financial loss and operational costs associated with fraud disputes, improving the security of online marketplaces.

3. **Mobile Payments:** With the rise of mobile wallets and payment apps, this model can be deployed to secure mobile transactions. It can identify unusual payment patterns or device information that may indicate that a user's account has been hijacked.

4. **Insurance:** Insurance companies can adapt this model to detect fraudulent claims. By analysing claim data, the model can identify patterns that suggest a claim may be exaggerated or entirely false, saving the industry billions of dollars annually.

5. **Return and Refund Fraud Prevention:** Fraudsters may exploit return policies by sending back counterfeit or used products or claiming refunds without returning the item. Machine learning models can examine historical purchase patterns, return frequency, reason codes, and product condition reports to identify fraudulent claims. By deploying these models, e-

commerce businesses can reduce losses associated with fraudulent returns, protect genuine customers from policy misuse, and maintain operational efficiency in handling returns.

6. **Affiliate Marketing Fraud Detection:** E-commerce businesses often run affiliate programs where partners earn commissions for driving traffic or sales. Fraudsters exploit this system through fake clicks, bots, or self-purchase to earn illegitimate commissions. Machine learning models can detect anomalies in traffic sources, conversion rates, and click patterns to identify fraudulent affiliates. Detecting such fraud ensures fair commission pay-outs and reduces the operational costs associated with invalid traffic and fake conversions.

7. SOFTWARE AND HARDWARE REQUIREMENTS

SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10
- **Programming Language:** Python 3.13.7
- **Development Environment:** Jupyter Notebook / VS Code
- **Libraries and Frameworks:**
    - **XGBoost** (core algorithm for fraud detection)
    - **scikit-learn** (for preprocessing, evaluation, and model utilities)
    - **NumPy** (for numerical computations)
    - **Pandas** (for data manipulation and analysis)
    - **Matplotlib, Seaborn** (for visualization of data and model results)
    - **imbalanced-learn** (for SMOTE or resampling techniques)
- **Package Management:** pip

HARDWARE REQUIREMENTS

- **Processor:** Intel Core i5 or AMD Ryzen 5
- **RAM:** 8 GB
- **Storage:** 20 GB free space (preferably SSD)
- **GPU:** Optional – NVIDIA GPU with CUDA support (for faster model training)